WORKED SOLUTION

STEP III 2025, Question 5

Pure •

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

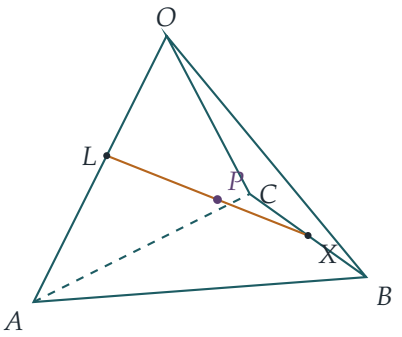
www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

Because O is above the plane of the non-collinear points A, B, C, the vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are linearly independent, so we may *compare coefficients* in any vector equation written in terms of them. Every point in the question is then found by writing it with unknown parameters and matching coefficients. Heights above the plane are just multiples of the coefficients, which makes "horizontal" easy to handle.

Solution

Since O is not in the plane ABC, no one of $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is a combination of the other two (if, say, $\mathbf{c} = p\mathbf{a} + q\mathbf{b}$ then O, A, B, C would be coplanar). So $x\mathbf{a} + y\mathbf{b} + z\mathbf{c} = x'\mathbf{a} + y'\mathbf{b} + z'\mathbf{c}$ forces $x = x'$, $y = y'$, $z = z'$. Note also that $k > 0$.



Part (i)

Write $\overrightarrow{OL} = \lambda\mathbf{a}$ and $\overrightarrow{OX} = (1-t)\mathbf{b} + t\mathbf{c}$. Since P lies on LX, for some s

$$\overrightarrow{OP} = (1-s)\lambda\,\mathbf{a} + s(1-t)\,\mathbf{b} + st\,\mathbf{c}.$$

Comparing with $\alpha\mathbf{a}+\beta\mathbf{b}+\gamma\mathbf{c}$:

$$(1-s)\lambda=\alpha, \qquad s(1-t)=\beta, \qquad st=\gamma.$$

Adding the last two gives $s=\beta+\gamma$, and then $t=\gamma/(\beta+\gamma)$, so

$$\overrightarrow{OX}=\frac{\beta\mathbf{b}+\gamma\mathbf{c}}{\beta+\gamma}.$$

Finally $1-s=1-\beta-\gamma=k+\alpha$, which is positive, so

$$\overrightarrow{OL}=\frac{\alpha}{1-s}\,\mathbf{a}=\frac{\alpha}{k+\alpha}\,\mathbf{a},$$

as required. (The coefficients were forced, so L and X are unique; and since $0<s<1$ and $0<t<1$, P lies strictly between L and X, with X strictly between B and C.)

Part (ii)

The same argument with the letters cycled ($\mathbf{a}\to\mathbf{b}\to\mathbf{c}\to\mathbf{a}$, $\alpha\to\beta\to\gamma\to\alpha$) gives

$$\overrightarrow{OM}=\frac{\beta}{k+\beta}\mathbf{b}, \quad \overrightarrow{ON}=\frac{\gamma}{k+\gamma}\mathbf{c}, \quad \overrightarrow{OY}=\frac{\gamma\mathbf{c}+\alpha\mathbf{a}}{\gamma+\alpha}, \quad \overrightarrow{OZ}=\frac{\alpha\mathbf{a}+\beta\mathbf{b}}{\alpha+\beta}.$$

Ask yourself How can we tell whether a plane through three given points is horizontal?

It is horizontal exactly when the three points are at the same height. Let O be at height $h>0$ above the plane ABC. Then A, B, C are each a vertical distance h below O, so the point $\mu\mathbf{a}$ on OA is a distance μh below O, and similarly on OB and OC. So L, M, N (which are not collinear, lying on three different edges away from O) are at depths

$$\frac{\alpha h}{k+\alpha}, \qquad \frac{\beta h}{k+\beta}, \qquad \frac{\gamma h}{k+\gamma}$$

below O. Now

$$\frac{\alpha}{k+\alpha}=\frac{\beta}{k+\beta} \iff \alpha k+\alpha\beta=\beta k+\alpha\beta \iff k(\alpha-\beta)=0 \iff \alpha=\beta,$$

using $k>0$. Similarly for the other pairs, so

$$\text{plane } LMN \text{ is horizontal} \iff \alpha=\beta=\gamma.$$

Next, where does OP meet the plane ABC? A point $x\mathbf{a}+y\mathbf{b}+z\mathbf{c}$ lies in that plane exactly when $x+y+z=1$ (it is then $\mathbf{a}+y(\mathbf{b}-\mathbf{a})+z(\mathbf{c}-\mathbf{a})$, and every point of the plane has this form). The point $\mu\overrightarrow{OP}$ has coefficient sum $\mu(\alpha+\beta+\gamma)$, so the line OP meets the plane at

$$\overrightarrow{OG}=\frac{\alpha\mathbf{a}+\beta\mathbf{b}+\gamma\mathbf{c}}{\alpha+\beta+\gamma}.$$

By comparing coefficients, this equals $\frac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c})$ if and only if

$$\frac{\alpha}{\alpha+\beta+\gamma}=\frac{\beta}{\alpha+\beta+\gamma}=\frac{\gamma}{\alpha+\beta+\gamma}=\frac{1}{3},$$

that is, if and only if $\alpha=\beta=\gamma$. Both conditions are equivalent to $\alpha=\beta=\gamma$, so the plane LMN is horizontal if and only if OP meets the plane ABC at the centroid G of triangle ABC. This completes the "show that".

In this case $\overrightarrow{OX}=\frac{1}{2}(\mathbf{b}+\mathbf{c})$, and likewise for Y and Z: the points X, Y, Z are the **midpoints** of BC, CA and AB respectively.

Part (iii)

We can write

$$\overrightarrow{OP}=k\,\mathbf{0}+\alpha\mathbf{a}+\beta\mathbf{b}+\gamma\mathbf{c},\qquad k+\alpha+\beta+\gamma=1,$$

with all four weights k,α,β,γ positive, precisely because $\alpha+\beta+\gamma<1$. So P is a weighted average of O, A, B, C with strictly positive weights, which means P lies strictly **inside** the tetrahedron $OABC$.

Put another way: positivity of α, β, γ alone places P inside the infinite three-sided cone at O spanned by the edges OA, OB, OC. The depth of P below O is $(\alpha+\beta+\gamma)h$, so the condition $\alpha+\beta+\gamma<1$ says that P is above the plane ABC, on the same side as O. (If the sum were 1, P would be on the face ABC; if greater than 1, beyond it.)

Checks

- Take $\alpha=\beta=\gamma=\frac{1}{4}$, so $k=\frac{1}{4}$: then $\overrightarrow{OL}=\frac{1}{2}\mathbf{a}$, $\overrightarrow{OM}=\frac{1}{2}\mathbf{b}$, $\overrightarrow{ON}=\frac{1}{2}\mathbf{c}$, and LMN is clearly the horizontal mid-section, as part (ii) predicts.
- As $k\to 0$ (so P approaches the face ABC), $\overrightarrow{OL}\to\mathbf{a}$: L moves down to A, which makes sense since L, P, X then all lie in the plane ABC.
- A computer check with 2000 random tetrahedra and random α,β,γ confirmed that P lies on LX (residual about 10^{-15}), and that LMN is horizontal exactly when G is the centroid.

Common slips

- Comparing coefficients without saying why it is allowed: it needs $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ to be independent, which is where "O lies above the plane" is used.
- Forgetting that $k>0$ is needed to cancel in $k(\alpha-\beta)=0$.
- Proving only one direction of the "if and only if".

Beyond STEP

The numbers $(k, \alpha, \beta, \gamma)$ are the *barycentric coordinates* of P with respect to the tetrahedron $OABC$: a point is inside exactly when all of them are positive, and on a face when one of them is zero. They are the natural coordinates for affine geometry and turn up again in linear algebra (affine combinations and convex hulls) and in computer graphics, where triangles are shaded by interpolating with barycentric weights. Strang's *Introduction to Linear Algebra* is a good place to see linear independence and coordinates used this way.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.